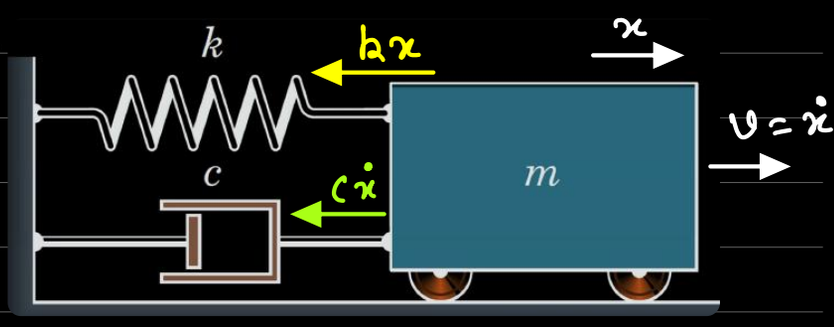
Consider the spring mass damper system shown in the figure.



Applying Laws of Motion to the system we get:

$$-kx - c\dot{x} = m\ddot{x} \Rightarrow \quad m\ddot{x} + c\dot{x} + kx = 0$$

$$\Rightarrow \quad \ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0$$

This conforms to standard differential equation governing Damped oscillations i.e.:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0 \quad (1) \text{ Where } \omega_0 = \sqrt{\frac{k}{m}},\ 2\beta = \frac{c}{m}$$

Here ω_0 would be the system's oscillation fuquency had there been no damping. To handle damped oscillations we shall learn to solve differential equations of form (1)

SECOND ORDER HOMOGENEOUS LINEAR ORDINARY D.E.:

An equation of the form $a\ddot{x} + b\dot{x} + cx = 0$ is called second order linear differential equation. To solve it we follow the following steps:

(1) Solve the corresponding characteristic equation: $a\lambda^2 + b\lambda + c = 0$

(2) If solutions λ_1 & λ_2 are real and distinct then:

$$\boxed{x(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}}$$

(3) If roots are repeated then:

$$\boxed{x(t) = Ae^{\lambda t} + Bte^{\lambda t}}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

④ If λ_1 & λ_2 are conjugate pairs i.e $\lambda_1 = -\beta + i\omega$ and $\lambda_2 = -\beta - i\omega$ then:

$$x(t) = e^{-\beta t}\left[A\cos\omega t + B\sin\omega t\right]$$

or $\boxed{x(t) = Ae^{-\beta t}\cos(\omega t + \phi)}$ ← Oscillatory Solution

This form of solution will be obtained when damping is not very large and our main focus will be on this form for JEE purposes.

For example for our standard DHMDE:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 = 0$$, characteristic equation will be:

$$\lambda^2 + 2\beta\lambda + \omega_0^2 = 0 \Rightarrow \lambda = \frac{-2\beta \pm \sqrt{4\beta^2 - 4\omega_0^2}}{2}$$

$$\Rightarrow \lambda_1 = -\beta + \sqrt{\beta^2 - \omega_0^2} \text{ and } \lambda_2 = -\beta - \sqrt{\beta^2 - \omega_0^2}$$

If β is sufficiently small, the roots will be complex and our required solution will then be:

$\boxed{x(t) = Ae^{-\beta t}\cos(\omega t + \phi)}$ where $\boxed{\omega = \sqrt{\omega_0^2 - \beta^2}}$

Here A and ϕ are arbitrary constants depending on initial conditions (initial velocity and initial position).

Examples: Solve the following ODEs:

(a) $\ddot{x} + 3\dot{x} + 2x = 0$ (b) $\ddot{x} + 2\dot{x} + x = 0$

(c) $\ddot{x} + 4\dot{x} + 5x = 0$

YOUTUBE.COM/@THEPHYSICSSHELTER

Solutions: (a) $\ddot{x} + 3\dot{x} + 2x = 0 \Rightarrow \lambda^2 + 3\lambda + 2 = 0$

$\Rightarrow (\lambda+1)(\lambda+2) = 0 \Rightarrow \lambda_1 = -1, \lambda_2 = -2$

$\Rightarrow x = Ae^{-t} + Be^{-2t}$ [∵ Real & Distinct Roots].

(b) $\ddot{x} + 2\dot{x} + x = 0 \Rightarrow \lambda^2 + 2\lambda + 1 = 0 \Rightarrow (\lambda+1)^2 = 1$

$\Rightarrow \lambda = -1$ [Repeated Root] $\Rightarrow x = Ae^{-t} + Bte^{-t}$

(c) $\ddot{x} + 4\dot{x} + 5 = 0 \Rightarrow \lambda^2 + 4\lambda + 5 = 0 \Rightarrow \lambda = \frac{-4 \pm 2i}{2}$

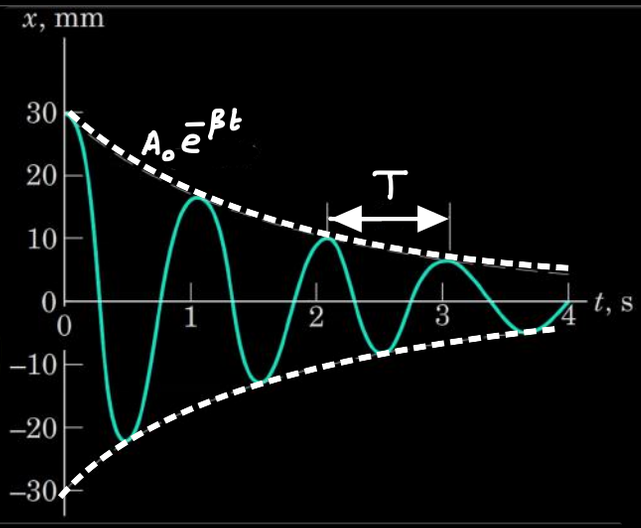
$\Rightarrow \lambda_1 = -2 - i$, $\lambda_2 = -2 + i \Rightarrow x(t) = Ae^{-2t}\cos(t + \phi)$

Plot Of Solution To DHMDE:

Recall that solution to $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x$ is given as:

$x = A_0 e^{-\beta t}\cos(\omega t + \phi)$ where $\omega = \sqrt{\omega_0^2 - \beta^2}$ ①

A typical plot of this is as shown in the adjacent figure with $m = 1\,kg$, $k = 36\,N/m$ and $c = 1\,Ns/m$.



[i.e $\beta = \frac{c}{2m} = 0.5\,\frac{Ns}{kgm}$ $\omega_0 = \sqrt{\frac{k}{m}} = 6\,rads^{-1}$]

Also @ $t = 0$, $x = 30\,mm$, $\dot{x} = 0$

→ Here Amplitude of Oscillation decays periodically

→ Strictly speaking motion is not periodic but we can consider

YOUTUBE.COM/@THEPHYSICSSHELTER

peak to peak time as some kind of a time period (which corresponds to time period of sinusoidal part of the solution).

According to equation ① it can be easily seen that time period of oscillation is given as:

$$T = \frac{2\pi}{\sqrt{\omega_0^2 - \beta^2}} = \frac{2\pi}{\omega}$$

It can be easily seen that if damping is small, time period is almost same as that of simple harmonic motion and increases with damping.

→ It can be seen that Amplitude of oscillation decays exponentially as $A = A_0 e^{-\beta t}$. Thus after every time period T, the amplitude decreases in a geometric progression i.e $\frac{A(t)}{A(t+T)} = e^{\beta T}$

DAMPING DECREMENT: The factor by which amplitude decreases in one time period is called damping decrement $e^{\beta T}$

LOGARITHMIC DECREMENT: Logarithm of damping decrement is known as logarithmic decrement. Mathematically:

$$\lambda = \ln(e^{\beta T}) \Rightarrow \boxed{\lambda = \beta T}$$ ← Logarithmic Decrement.

Using logarithmic decrement we can write the law of diminishing amplitude as:

$$\boxed{A = A_0 e^{-\lambda t/T}}$$

Time Constant: The time in which amplitude of oscillation decays to 1/e times the initial value is called time constant τ.

YOUTUBE.COM/@THEPHYSICSSHELTER

Mathematically, $A = A_0 e^{-\beta\tau} = A_0 e^{-1} \Rightarrow \boxed{\tau = 1/\beta = T/\lambda}$

Number of Oscillations in one time constant : $\boxed{N_e = \frac{\tau}{T} = \frac{1}{\lambda}}$

QUALITY (DAMPED OSCILLATIONS): $\boxed{Q = \frac{\pi}{\lambda}}$

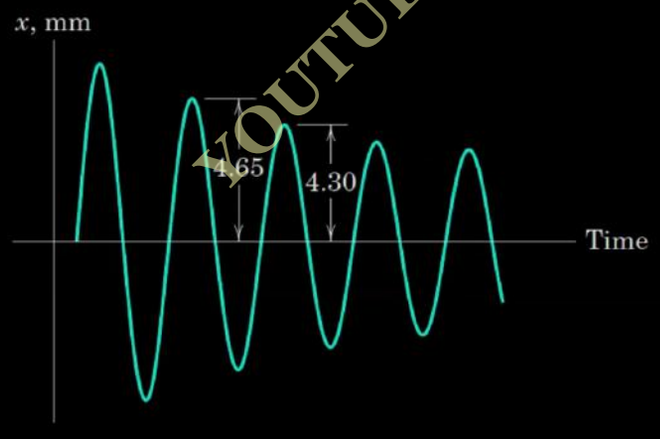
Quality is a measure of number of oscillations a system makes during one time constant (or by the time amplitude becomes 1/e times).

Energy in Damped Oscillations: Recall that energy of an oscillator is proportional to square of it amplitude.

$\therefore \because A = A_0 e^{-\beta t} \Rightarrow \boxed{E = E_0 e^{-2\beta t}}$

Example:

8/32 A linear harmonic oscillator having a mass of 1.10 kg is set into motion with viscous damping. If the frequency is 10 Hz and if two successive amplitudes a full cycle apart are measured to be 4.65 mm and 4.30 mm as shown, compute the viscous damping coefficient c.



Solution: $\frac{c}{m} = 2\beta \Rightarrow c = 2\beta m$

Now $A = A_0 e^{-\beta T} = A_0 e^{-\beta/\nu}$

$\Rightarrow \beta = \nu \ln(A_0/A)$

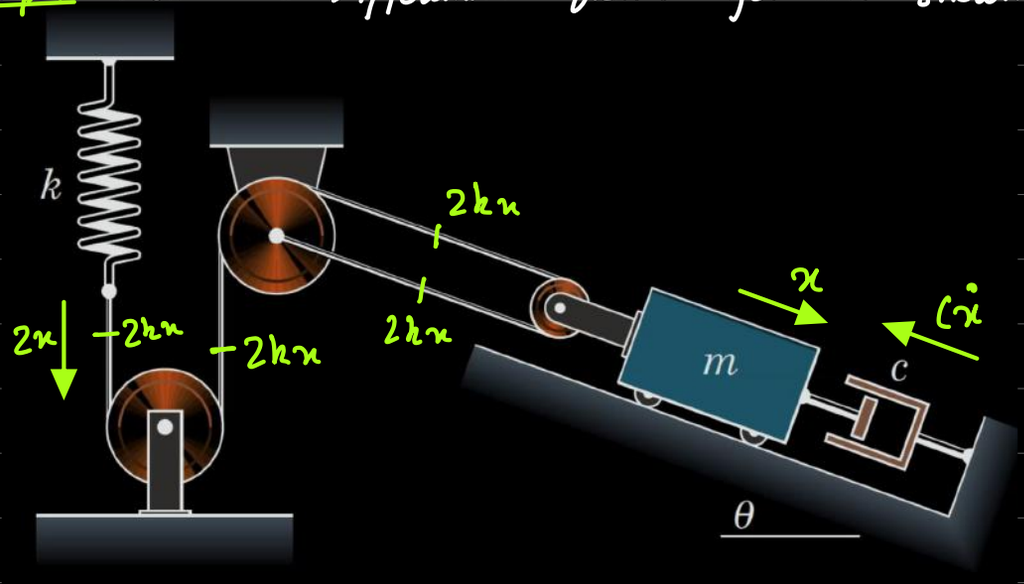
$\Rightarrow c = 2\nu \ln(A_0/A) \times m$

$= 2 \times 10 \ln(4.65/4.3) \times 1.10$

$\Rightarrow C = 1.72$ Ns/m Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

Example Devin the Differential Equation for the shown system.



Method I: Force Method: We shall consider changes in forces from mean position (∵ net force in mean position was zero ∴ net change in force must also be net force.

$$-4kx - c\dot{x} = m\frac{d^2x}{dt^2} \Rightarrow m\ddot{x} + c\dot{x} + 4kx = 0$$

$$T = \frac{2\pi}{\sqrt{\omega_0^2 - \beta^2}} \quad \text{where } \beta = \frac{c}{2m},\ \omega_0 = \sqrt{\frac{4k}{m}}$$

Method 2: Energy Method: Rate of work done by dissipative forces must be equal to rate of change of mechanical energy of the system.

Here it is more convenient to consider x from natural length configuration of the system.

$$\Rightarrow -c\dot{x}^2 = \frac{d}{dt}\left(-mgx\sin\theta + \frac{1}{2}k(2x)^2 + \frac{1}{2}m(\dot{x})^2\right)$$

$$\Rightarrow -c\dot{x}^2 = -mg\sin\theta\,\dot{x} + 4kx\dot{x} + m\dot{x}\ddot{x}$$

Divide by $\dot{x}$ and rearrange:

YOUTUBE.COM/@THEPHYSICSSHELTER

$$m\ddot{x} + C\dot{x} + 4kx - mg\sin\theta = 0$$

$$\Rightarrow m\ddot{x} + C\dot{x} + 4k\left(x - \frac{mg\sin\theta}{4k}\right) = 0$$

Here $\frac{mg\sin\theta}{4k}$ represents mean value of x,

we can use variable substitution, $y = x - \frac{mg\sin\theta}{4k}$

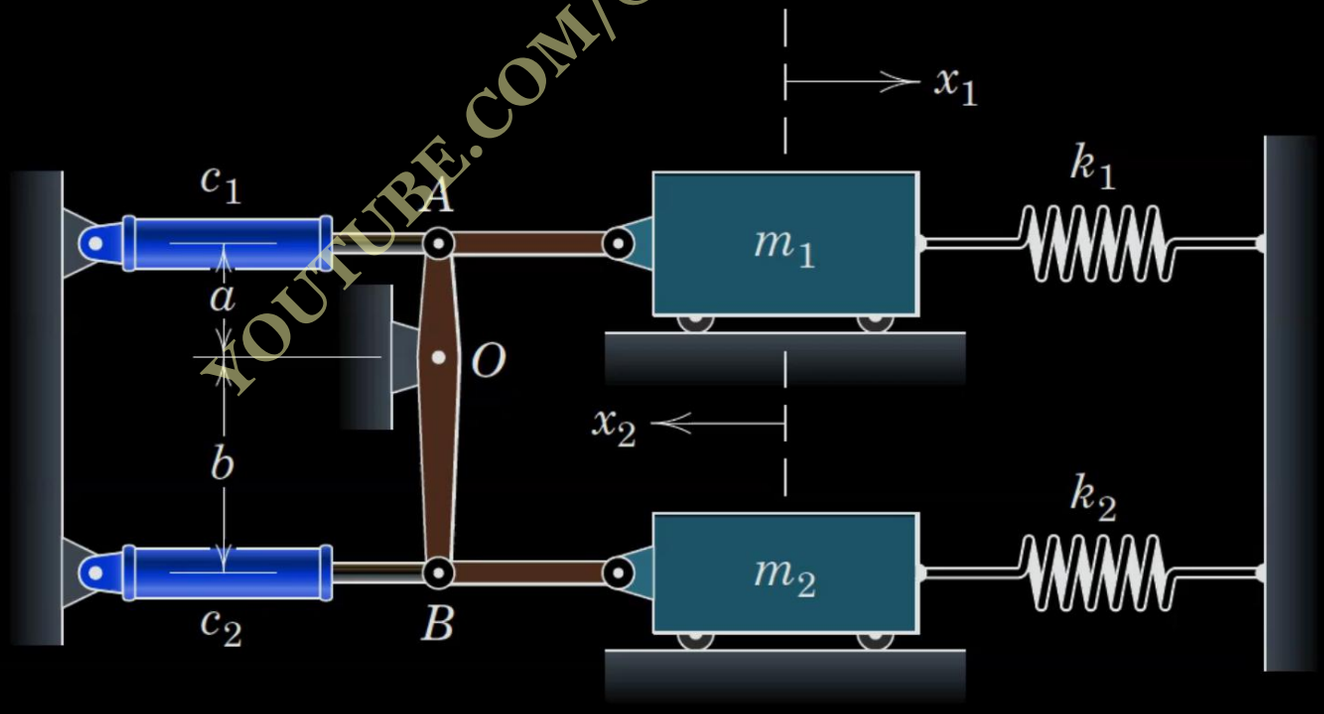
$$\Rightarrow \dot{y} = \dot{x} \quad \& \quad \ddot{y} = \ddot{x}$$

$$\Rightarrow m\ddot{y} + C\dot{y} + 4ky = 0$$ This is same equation as before.

(Note that constant term does not affect time period).

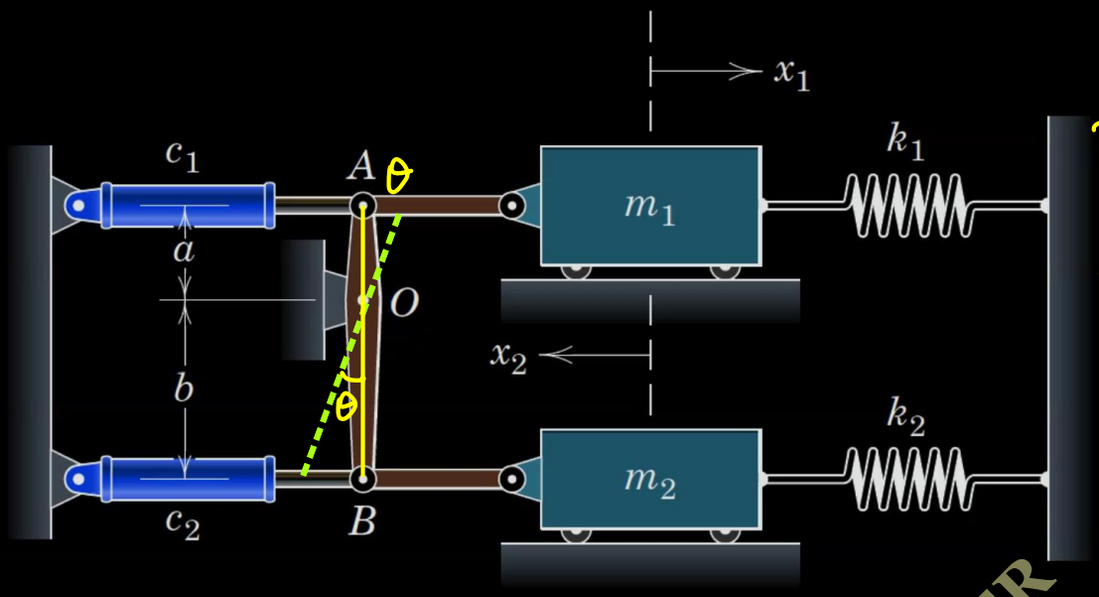
Example:

8/35 Derive the differential equation of motion for the system shown in terms of the variable x_1. Neglect friction and the mass of the linkage.



YOUTUBE.COM/@THEPHYSICSSHELTER

Solution:



$x_1 = a\theta$

$x_2 = b\theta$

$\Rightarrow x_2 = \frac{b}{a} x_1$

Here we shall use Energy method:

$$-c_1 \dot{x}_1^2 - c_2 \dot{x}_2^2 = \frac{d}{dt}\left(\frac{1}{2} m_1 \dot{x}_1^2 + \frac{1}{2} k_1 x_1^2 + \frac{1}{2} m_2 \dot{x}_2^2 + \frac{1}{2} k_2 x_2^2\right)$$

Now $x_2 = \frac{b}{a} x_1 \Rightarrow \dot{x}_2 = \frac{b}{a}\dot{x}_1$, $\ddot{x}_2 = \frac{b}{a}\ddot{x}_1 = \mu \ddot{x}_1$

where $\mu = b/a$.

$$\Rightarrow -\dot{x}_1^2 (c_1 + \mu^2 c_2) = \frac{d}{dt}\left[\frac{\dot{x}_1^2}{2}\left[m_1 + \mu^2 m_2\right] + \frac{x_1^2}{2}\left[k_1 + \mu^2 k_2\right]\right]$$

$$\Rightarrow -\dot{x}_1^2 (c_1 + \mu^2 c_2) = \dot{x}_1 \ddot{x}_1 \left[m_1 + \mu^2 m_2\right] + x_1 \dot{x}_1 \left[k_1 + \mu^2 k_2\right]$$

$$\Rightarrow \left(m_1 + \mu^2 m_2\right)\ddot{x}_1 + \left(c_1 + \mu^2 c_2\right)\dot{x}_1 + \left(k_1 + \mu^2 k_2\right) x_1 = 0$$

EXAMPLE : A certain oscillator follows the differential equation:

$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$, at $t = 0$, $x = x_0$ and $\dot{x} = v_0$, find A and ϕ, if solution is given by:

$$x = A e^{-\beta t} \cos(\omega t + \phi)$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Sol: $x(0) = x_0 \Rightarrow A\cos\phi = x_0$ ①

$$\dot{x} = -A\omega e^{-\beta t}\sin(\omega t+\phi) - A\beta e^{-\beta t}\cos(\omega t+\phi)$$

$$\Rightarrow \dot{x}(0) = -A\omega\sin\phi - A\beta\cos\phi = V_0 \quad ②$$

$$\Rightarrow -\omega A\sqrt{1-\left(\frac{x_0}{A}\right)^2} - \beta x_0 = V_0$$

$$\Rightarrow A^2 - x_0^2 = \left(\frac{V_0+\beta x_0}{\omega}\right)^2$$

$$\Rightarrow A = \sqrt{\left(\frac{V_0+\beta x_0}{\omega}\right)^2 + x_0^2} \quad \text{Ans.}$$

$$\cos\phi = \frac{x_0}{A}$$

If $x_0 > 0$, $V_0 > 0$, $\phi \in$ fourth quadrant

If $x_0 > 0$, $V_0 < 0$, $\phi \in$ first Quadrant

If $x_0 < 0$, $V_0 > 0$, $\phi \in$ Third Quadrant

If $x_0 < 0$, $V_0 < 0$, $\phi \in$ Second Quadrant

YOUTUBE.COM/@THEPHYSICSSHELTER

IRODOV:

4.69. A point performs damped oscillations with frequency ω and damping coefficient β according to the law (4.1b). Find the initial amplitude a_0 and the initial phase ϕ if at the moment $t = 0$ the displacement of the point and its velocity projection are equal to

(a) $x(0) = 0$ and $v_x(0) = \dot{x}_0$;
(b) $x(0) = x_0$ and $v_x(0) = 0$.

(4.1b): $x = A_0 e^{-\beta t} \cos(\omega t + \phi)$

Sol: (a) $x = A_0 e^{-\beta t} \cos(\omega t + \phi)$, $x(0) = 0$

$\Rightarrow 0 = A_0 \cos\phi$ (1), Also $\dot{x} = A_0 e^{-\beta t}[-\beta\cos(\omega t+\phi) - \omega \sin(\omega t+\phi)]$

Now $\dot{x}(0) = \dot{x}_0 \Rightarrow A_0[-\beta\cos\phi - \omega\sin\phi] = \dot{x}_0$ (2)

Using (1) in (2): $-A_0\omega\sin\phi = \dot{x}_0 = -A_0\omega\sqrt{1-\cos^2\phi} = \dot{x}_0$

$\Rightarrow \boxed{A_0 = \left|\frac{\dot{x}_0}{\omega}\right|}$ Ans. Also if $\dot{x}_0 > 0$, $\Rightarrow \sin\phi < 0$

$\Rightarrow \phi = -\pi/2$

and if $\dot{x}_0 < 0 \Rightarrow \sin\phi > 0$, $\Rightarrow \phi = +\pi/2$. Ans.

(b) $x(0) = x_0$, $\dot{x}(0) = 0$, $x = A_0 e^{-\beta t}\cos(\omega t+\phi)$

$\Rightarrow x_0 = A_0\cos\phi$ (1), $\dot{x} = A_0 e^{-\beta t}[-\beta\cos(\omega t+\phi) - \omega\sin(\omega t+\phi)]$

$\dot{x}(0) = A_0[-\beta\cos\phi - \omega\sin\phi] = 0$ (2)

Now using (1) in (2): $-\beta x_0 - A_0\omega\sqrt{1-(x_0/A_0)^2} = 0$

$\Rightarrow \omega\sqrt{A_0^2 - x_0^2} = -\beta x_0 \Rightarrow A_0^2 = \left(\frac{\beta x_0}{\omega}\right)^2 + x_0^2$

$\Rightarrow \boxed{A_0 = |x_0|\sqrt{1+(\beta/\omega)^2}}$ Ans [To keep positive Amplitude]

Also from (2), $\tan\phi = -\beta/\omega \Rightarrow \phi = \tan^{-1}(-\beta/\omega)$ if $x_0 > 0$

or $\phi = \pi + \tan^{-1}(-\beta/\omega)$ if $x_0 < 0$

YOUTUBE.COM/@THEPHYSICSSHELTER
AMBARISH SIR
https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

IRODOV:

4.70. A point performs damped oscillations with frequency $\omega = 25\ s^{-1}$. Find the damping coefficient β if at the initial moment the velocity of the point is equal to zero and its displacement from the equilibrium position is $\eta = 1.020$ times less than the amplitude at that moment.

Solution: Refer to previous problem, $A_0 = x_0\sqrt{1+(\beta/\omega)^2}$

$$\Rightarrow \quad \beta = \omega\sqrt{\left(\frac{A_0}{x_0}\right)^2 - 1} = \omega\sqrt{\eta^2 - 1} \quad \text{ans.}$$

[Note that ω should be converted to radians per second]

IRODOV EXAMPLE:

4.71. A point performs damped oscillations with frequency ω and damping coefficient β. Find the velocity amplitude of the point as a function of time t if at the moment $t = 0$

(a) its displacement amplitude is equal to a_0;

(b) the displacement of the point $x(0) = 0$ and its velocity projection $v_x(0) = \dot{x}_0$.

Sol: (a) $\quad x = A_0 e^{-\beta t}\cos(\omega t + \phi) \Rightarrow v = -A_0 e^{-\beta t}[\beta\cos(\omega t+\phi) + \omega\sin(\omega t+\phi)]$

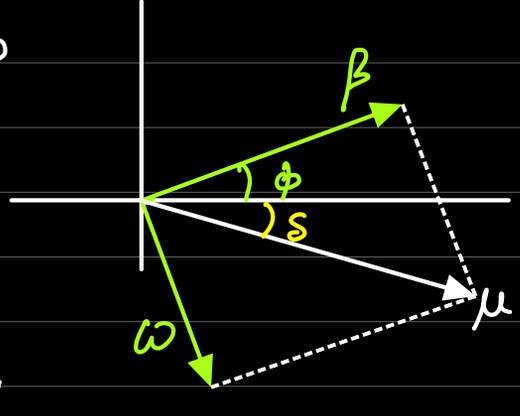
Now the term in the bracket is sum of two phasors at 90°

($\because \sin(\omega t+\phi) = \cos(\omega t + \phi - \pi/2)$. Adding the two phasors:

$$\mu = \sqrt{\omega^2+\beta^2}, \quad \delta = \tan^{-1}(\omega/\beta) - \phi$$

$$\Rightarrow v = -A_0 e^{-\beta t}[\mu\cos(\omega t + \delta)]$$

$$\Rightarrow \text{Amplitude} = \boxed{A_0\sqrt{\omega^2+\beta^2}\, e^{-\beta t}} \quad \text{ans.}$$



(b) For this case, $x = A_0 e^{-\beta t}\cos(\omega t + \phi)$ where $A_0 = \frac{\dot{x}_0}{\omega}$

(as found in 4.69).

YOUTUBE.COM/@THEPHYSICSSHELTER

Now $\frac{dx}{dt} = -A_0 e^{-\beta t}[\beta\cos(\omega t+\phi) + \omega \sin(\omega t+\phi)]$, Now the term is bracket is the same term that we evaluated using phasor addition for part (a) thus velocity amplitude:

$$\frac{\dot{x}_0}{\omega}\sqrt{\omega^2+\beta^2}\; e^{-\beta t} = \dot{x}_0\sqrt{1+(\beta/\omega)^2}\, e^{-\beta t} \quad \text{Ans.}$$

EXAMPLE:

4.73. A mathematical pendulum oscillates in a medium for which the logarithmic damping decrement is equal to $\lambda_0 = 1.50$. What will be the logarithmic damping decrement if the resistance of the medium increases $n = 2.00$ times? How many times has the resistance of the medium to be increased for the oscillations to become impossible?

Solution: $\lambda_0 = \beta_0 T_0 = \beta_0 \times \frac{2\pi}{\sqrt{\omega_0^2 - \beta_0^2}}$ ①

Now if resistance is made n times,

$$\lambda = n\beta_0 \times \frac{2\pi}{\sqrt{\omega_0^2 - (n\beta_0)^2}} \quad ②$$

Eliminating β_0 from ① & ②:

$$\lambda = \frac{n\lambda_0}{\sqrt{1+(1-n^2)(\lambda_0/2\pi)^2}} \quad \text{Ans}$$

Now for non oscillatory solution, let damping increase by a factor of N. Then:

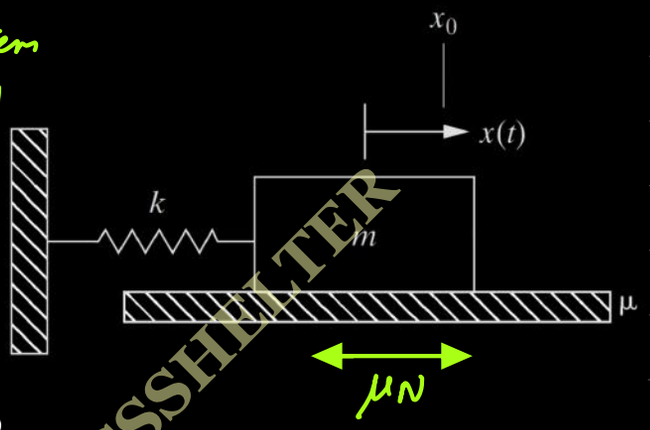
$$\omega_0^2 - (N\beta_0)^2 = 0 \Rightarrow N = \frac{\omega_0}{\beta_0}$$

From ①, $\beta_0 = \frac{\lambda_0\omega_0}{\sqrt{4\pi^2+\lambda_0^2}} \Rightarrow N = \sqrt{1+\left(\frac{2\pi}{\lambda_0}\right)^2}$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

COULOMB DAMPING: If instead of a viscous damping we have dry friction damping we call the situation as coulomb damping. In this case the amplitude decreases in Arithmetic progression until the spring force at one of the extremes becomes less than the limiting friction. This can be readily seen as follows:

Consider the spring mass dry friction system shown in the figure. The differential equation governing the system is:



$$-kx - \mu N = m\frac{d^2x}{dt^2} \quad \text{if } \dot{x} > 0$$

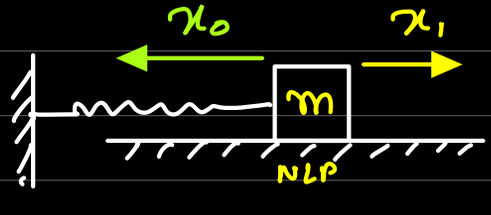
or $$m\frac{d^2x}{dt^2} + kx + \mu N = 0 \quad \text{if } \dot{x} > 0$$

This is clearly an SHM until $\dot{x} \geq 0$, now when the velocity reverses, the sign of friction changes and our differential equation becomes:

$$m\frac{d^2x}{dt^2} + kx - \mu N = 0$$ This also has a simple harmonic solution. Thus the solution to the entire physical situation is piecewise simple harmonic. (piecewise half cycles of SHM with alternating mean positions).

Now Suppose the block is released from x_0 on the left extreme and it reverses its motion at a distance x_1 at the other extreme position.

Then by work Energy theorem:



$$\frac{1}{2}kx_0^2 - \mu N(x_1 + x_0) = \frac{1}{2}kx_1^2$$

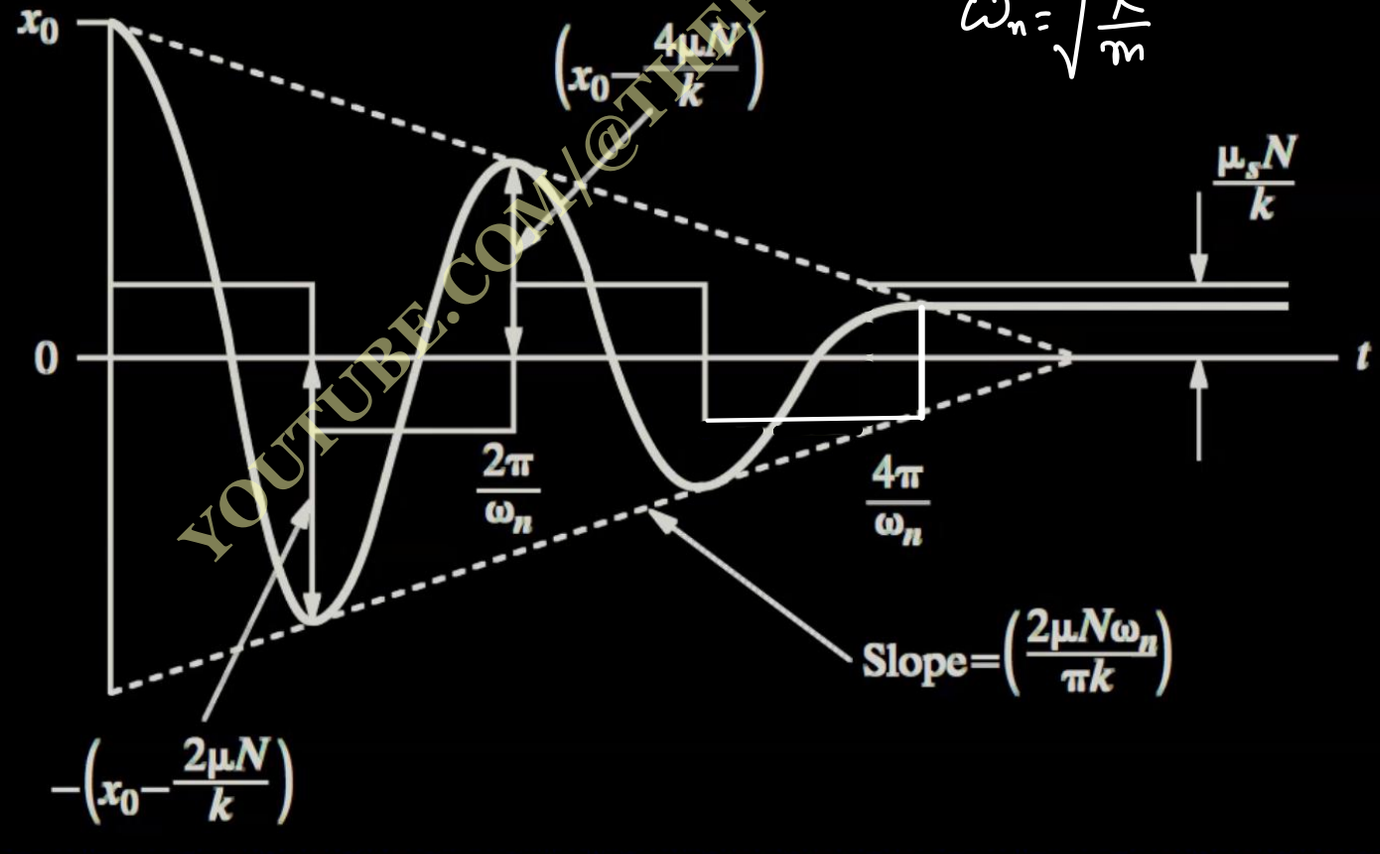
$$\Rightarrow \frac{1}{2}k(x_0^2 - x_1^2) = \mu N(x_1 + x_0) \Rightarrow x_0 - x_1 = \frac{2\mu N}{k}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

Thus we can see that amplitude has decreased by $\frac{2\mu N}{k}$ in one half cycle. Similarly $x_2 = x_1 - 2\mu N/k$ (when block returns to the left extreme) and this pattern continues until limiting friction becomes large enough to overcome the spring force at the extreme position. Thus we can calculate the number of half cycles as: $kx_n \leq \mu N \Rightarrow k\left(x_0 - 2\frac{\mu N}{k} \times n\right) \leq \mu N$

$$\Rightarrow \quad 2n\mu N \geq kx_0 - \mu N \quad \text{or} \quad n \geq \left(\frac{kx_0}{2\mu N} - 1\right)$$

However, n is constrained to be an integer thus $n = \text{GIF}\left[\frac{kx_0}{2\mu N}\right]$



YOUTUBE.COM/@THEPHYSICSSHELTER

→ Note that final PE in the spring is not zero in general (Finally the block will stop at the dotted line in the graph.

EXAMPLE:

4.82. A bar of mass $m = 0.50$ kg lying on a horizontal plane with a friction coefficient $\mu = 0.10$ is attached to the wall by means of a horizontal non-deformed spring. The stiffness of the spring is equal to $\varkappa = 2.45$ N/cm, its mass is negligible. The bar was displaced so that the spring was stretched by $x_0 = 3.0$ cm, and then released. Find:

(a) the period of oscillation of the bar;

(b) the total number of oscillations that the bar performs until it stops completely.

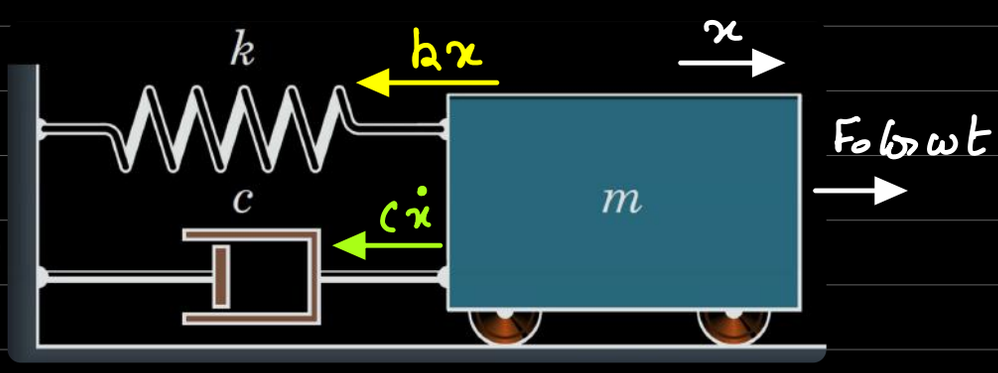
Sol: (a) $T = 2\pi\sqrt{\frac{m}{\varkappa}} = 2\pi\sqrt{\frac{0.5}{245}} = 0.28$ sec.

(b) Now no of half cycles: $\text{GIF}\left[\frac{k x_0}{2\mu N}\right]$

$= \text{GIF}\left[\frac{245 \times 0.03}{2 \times 0.1 \times 5}\right]$ $= \text{GIF}\left[\frac{147}{20}\right] = 7$ Ans.

YOUTUBE.COM/@THEPHYSICSSHELTER

FORCED OSCILLATIONS:



Consider the system shown in the figure. The governing Differential Equation for the system:

$$F_0 \cos\omega t - kx - c\dot{x} = m\ddot{x}$$

$$\Rightarrow m\ddot{x} + c\dot{x} + kx = F_0 \cos\omega t$$, We can rearrange this equation as:

$$\ddot{x} + (c/m)\dot{x} + (k/m)x = (F_0/m)\cos\omega t$$

This equation is the standard equation of Forced Simple Harmonic Motion its form is:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \quad (1)$$

where, $2\beta = c/m$, $\omega_0 = \sqrt{k/m}$ and $f_0 = F_0/m$.

The "steady state" solution of equation (1) is given by

$$x = A\cos(\omega t - \phi) \quad (2)$$ where, $$\boxed{A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}} \quad (3)$$

and $$\boxed{\phi = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)} \quad (4)$$

and the complete solution is the superposition of the the steady state solution given by (2) and the solution of the allied damped oscillations equation i.e:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0.$$

The steady solution as defined by equations (3) & (4) above can be easily proved using the complex number approach (or complex phasors) however here we shall take a more tedious trigonometric approach.

YOUTUBE.COM/@THEPHYSICSSHELTER

Proof Of The Solution Of The Forced Damped Oscillator:

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos\omega t \quad (1)$$

Let us assume a solution of the form $x = A\cos(\omega t - \phi)$, our task is to evaluate A and ϕ.

Now: $\dot{x} = -A\omega \sin(\omega t + \phi)$, $\ddot{x} = -A\omega^2 \cos(\omega t - \phi)$

Substituting $\dot{x}$ and $\ddot{x}$ in eqn (1) we get:

$$A(\omega_0^2 - \omega^2)\cos(\omega t - \phi) - 2\beta\omega \sin(\omega t - \phi) = f_0 \cos(\omega t) \quad (2)$$

Now Recall that: $\cos(\omega t - \phi) = \cos\omega t \cos\phi + \sin\omega t \sin\phi$

and $\sin(\omega t - \phi) = \sin\omega t \cos\phi - \cos\omega t \sin\phi$.

We shall use these substitutions in equation (2) and equate the coefficients of Sines and Cosines on LHS and RHS (since the equation (2) is identically true for all times).

Thus we get:

$$A[(\omega_0^2 - \omega^2)\cos\phi + 2\beta\omega \sin\phi] = f_0 \quad (3)$$

and

$$A[(\omega_0^2 - \omega^2)\sin\phi - 2\beta\omega\cos\phi] = 0 \quad (4)$$

(4)/(3) gives:

$$\frac{(\omega_0^2 - \omega^2)\tan\phi - 2\beta\omega}{(\omega_0^2 - \omega^2) + 2\beta\omega\tan\phi} = 0$$

$$\Rightarrow \tan\phi = \left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right) \Rightarrow \phi = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

Now substituting this in (3), we get:

$$A = f_0 / \sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

We can multiply both numerator and denominator of above equation to get an expression similar to NCERT notation:

$$\boxed{A = \frac{F_0}{[m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2]^{1/2}}}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

NCERT also gives an expression of ϕ equivalent to:

$$\boxed{\tan\phi = \left(\frac{v_0}{\omega x_0}\right)}$$

This can be derived trivially from our solution equation itself:

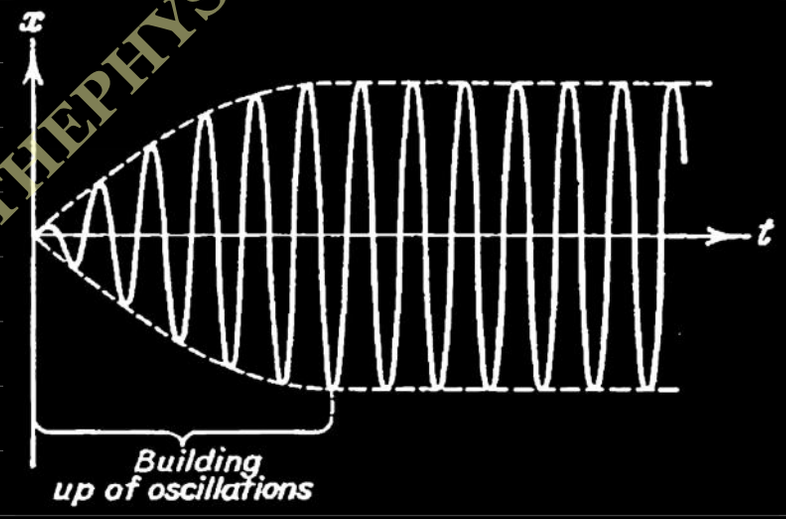
$$x = A\cos(\omega t - \phi) \Rightarrow x_0 = A\cos\phi \;,\; v = -A\omega\sin(\omega t - \phi)$$
$$\Rightarrow v_0 = A\omega\sin\phi$$
$$\Rightarrow \tan\phi = \frac{v_0}{x_0\omega}.$$

$x_0 \rightarrow$ Initial Position,
$v_0 \rightarrow$ Initial Velocity.

→ Note NCERT has taken a solution of the form $A\cos(\omega t + \phi)$ instead of $A\cos(\omega t - \phi)$ so in their expression for $\tan\phi$ they have a negative sign.

PLOT OF TIME RESPONSE OF FORCED OSCILLATIONS: ($x_0 = 0$)

The complete solution of Forced Oscillaton is superposition of the "transient response" [Exponentially decaying response of allied DHMDE and steady state solution]. Since the transient response is exponentially decaying, after a long time the initial conditions of the system do not matter for the steady state response.



RESONANCE IN FORCED OSCILLATIONS: Recall that the amplitude of oscillations in forced damped systems is given as:

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}},$$ thus the amplitude

of oscillations depends on the driving frequency ω.

YOUTUBE.COM/@THEPHYSICSSHELTER

The driving frequency ω at which the amplitude becomes maximum is known as resonance frequency, and such a driven system is said to be in resonance. Now for max amplitude, we need to minimise the term in the square root in the denominator. Thus for A_{max}:

$$\frac{d}{d\omega}\left[(\omega_0^2-\omega^2)^2+4\beta^2\omega^2\right]=0$$

$$\Rightarrow -4(\omega_0^2-\omega^2)\omega+8\beta^2\omega=0 \Rightarrow \omega=0, \text{ or } \pm\sqrt{\omega_0^2-2\beta^2}$$

$\omega=0$ corresponds to a maximum and ω can't be negative thus we get:

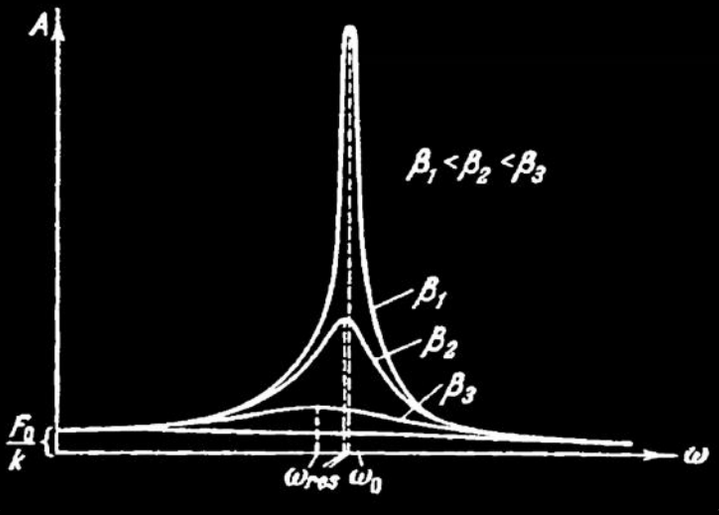
$$\boxed{\omega_{res}=\sqrt{\omega_0^2-2\beta^2}} \quad (1)$$

and the corresponding amplitude of oscillation is given by:

$$\boxed{A_{res}=\frac{f_0}{2\beta\sqrt{\omega_0^2-\beta^2}}} \quad (2)$$

FREQUENCY DEPENDENCE OF AMPLITUDE:

From equation (2) above it is clear that if damping is absent, the oscillation amplitude would go to infinity. The adjacent graph represents variation of oscillation amplitude with changing damping coefficient and varying driving frequency.



→ It can also be seen that for small damping, the maximum amplitude occurs near the natural frequency (undamped frequency) of the system. In this case neglecting β^2 terms in amplitude,

$$\boxed{A=f_0/(\omega^2-\omega_0^2)}$$ ← Small damping ω far from ω_0.

YOUTUBE.COM/@THEPHYSICSSHELTER

[Recall that $A = f_0 / \sqrt{(\omega^2 - \omega_0^2)^2 + 4\beta^2\omega^2}$].

→ It can also be seen readily from equation ① that resonance is not possible if $2\beta^2 > \omega_0^2$. In such cases the amplitude diminishes monotonically with increasing driving frequency.

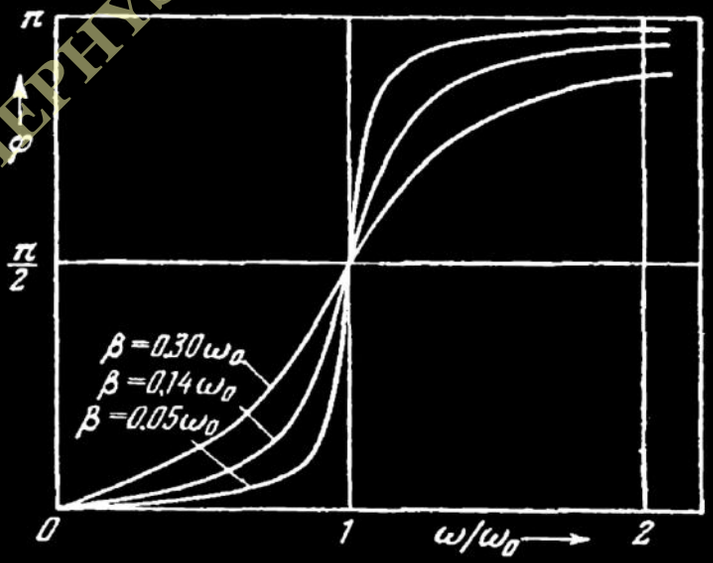
→ If ω is very close to ω_0, we may ignore $(\omega^2 - \omega_0^2)$ term in the denominator of amplitude expression.

$$\Rightarrow \boxed{A = \frac{f_0}{2\beta\omega} = \frac{F_0}{c\omega}}$$

Variation of Phase Angle With Driver Frequency:

$x = A\cos(\omega t - \phi)$

The adjacent graph shows variation of phase angle with driving frequency and damping coefficient. It is worth observing the following.



→ At very low driving frequency the system oscillates nearly in the same phase as the driving force (with some phase lag though).

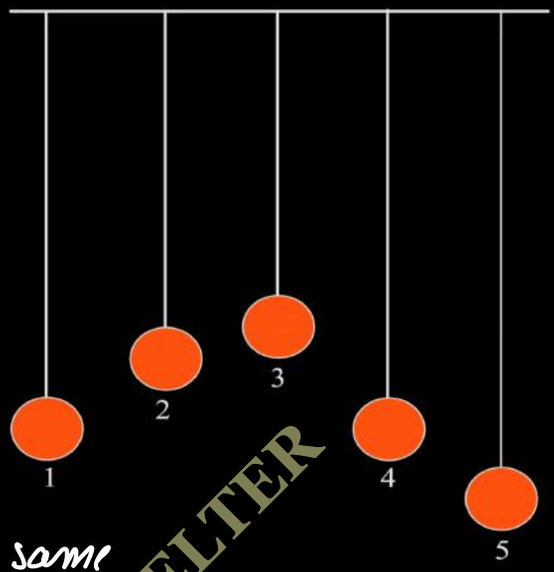
→ At very large driving frequencies, the displacement is nearly π out of phase (strictly lagging) with the driving force which means when the force is maximum to the right, the block will be at the left extreme position.

→ At natural frequency (undamped oscillator sym), the displacement of the system lags behind the forcing function by a phase of $\pi/2$.

YOUTUBE.COM/@THEPHYSICSSHELTER

→ Increasing the damping increases the phase lag. ($\omega < \omega_0$)

NCERT EXAMPLE: In the shown situation, pendulum 1 is given some push as a result of which, the support begins to provide a periodic force to the strings of other pendulums also. Which pendulum out of 2, 3, 4 and 5 will oscillate with maximum amplitude.



Sol: Since Pendulum 4 has same length as ① its natural frequency is same as the driving frequency and therefore it will go in to resonance and hence a large amplitude.

→ By same logic we can also say that ③ will have a smaller amplitude than ② because its natural frequency is farther from driving (①) frequency than that of ②.

Quality Factor: Recall that for small damping, amplitude is given as $A = \frac{F_0/m}{2\beta\omega_0}$ (Driving frequency ≈ natural frequency)

Now if a constant force F_0 were applied, the system would get displaced from its mean position by $x_0 = F_0/k$.

The ratio of A to x_0 in such a case is called quality factor.

$$Q = A/x_0 = \frac{\omega_0}{2\beta} = \frac{2\pi}{2\beta T} = \frac{\pi}{\lambda}$$

$$\Rightarrow \boxed{Q = \frac{\omega_0}{2\beta} = \frac{\pi}{\lambda}}$$

Here λ is the logarithmic decrement of corresponding damped system.

→ In Engineering Q is also known as amplitude ratio or magnification factor.

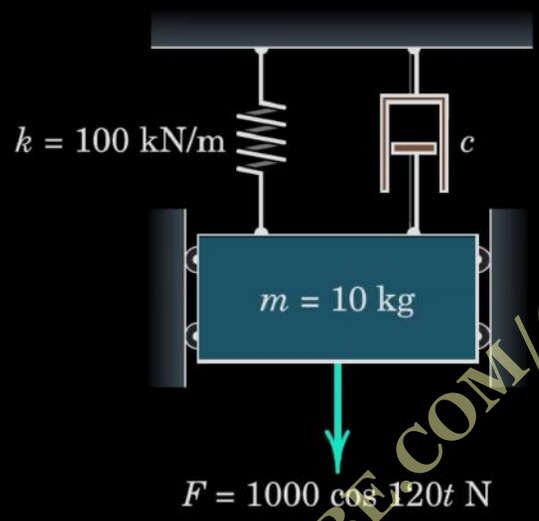
YOUTUBE.COM/@THEPHYSICSSHELTER

→ Note that throughout our analysis we had assumed $F = F_0 \cos(\omega t)$ however even if the forcing function were $F = F_0 \cos(\omega t + \delta)$ our solution will remain same in essence (δ just signifies the time when we started our observation).
Thus with $F = F_0 \cos(\omega t + \delta)$ our solution would become $x = A\cos(\omega t + \delta - \phi)$ where A and ϕ are still given by the expressions that we derived.

→ Thus if $F = F_0 \sin(\omega t)$, $x = A \sin(\omega t - \phi)$.

EXAMPLE :

Determine the amplitude X of the steady-state motion of the 10-kg mass if (a) $c = 500$ N·s/m and (b) $c = 0$.



Solution: (a)

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}$$

$f_0 = F_0/m = 1000/10 = 100$
$\omega_0 = \sqrt{k/m} = 100$
$\beta = c/2m = 25$

Putting the values: $A = 0.0134$ m
$\Rightarrow A = 1.34$ cm.

(b) Now $\beta = 0$ Putting the values in same formula, $A = 2.27$ cm.

YOUTUBE.COM/@THEPHYSICSSHELTER

EXAMPLE:

The 30-kg cart is acted upon by the harmonic force shown in the figure. If $c = 0$, determine the range of the driving frequency ω for which the magnitude of the steady-state response is less than 75 mm.



Sol: $A = \dfrac{f_0}{\sqrt{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}}$, Here $\beta = 0$

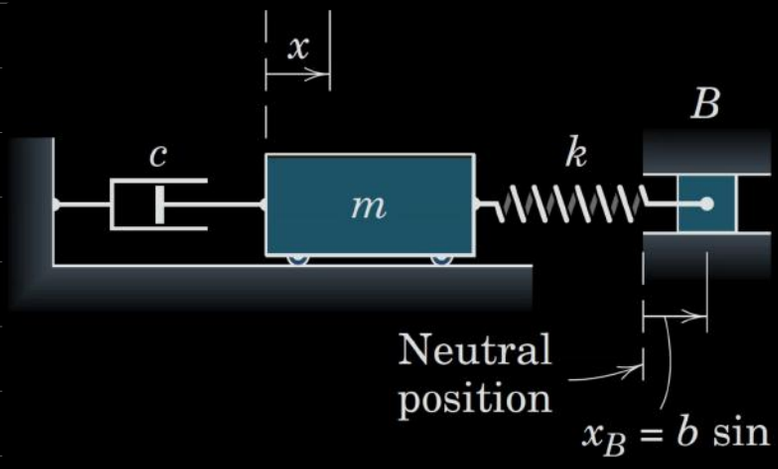
$\Rightarrow A = \left| \dfrac{f_0}{(\omega_0^2 - \omega^2)} \right| \leq 0.075$

Now $f_0 = F_0/m = 5/6$, $\omega_0 = \sqrt{k/m} = 6$

$\Rightarrow \left| \dfrac{5/6}{36 - \omega^2} \right| \leq 0.075 \Rightarrow \omega \leq 4.99 \text{ rad s}^{-1}$ or $\omega \geq 6.86 \text{ rad s}^{-1}$

YOUTUBE.COM/@THEPHYSICSSHELTER

BASE EXCITATION:



Sometimes instead of directly applying a sinusoidal force to the system, the end of the spring is given a sinusoidal displacement, this is equivalent to applying a force function:

$F = F_0 \sin(\omega t)$ where $F_0 = kb$. (b is the displacement amplitude of the other end of the spring).

This can be shown as follows:

Considering the FBD of the block:



$$-k(x - x_B) - c\dot{x} = m\ddot{x}$$

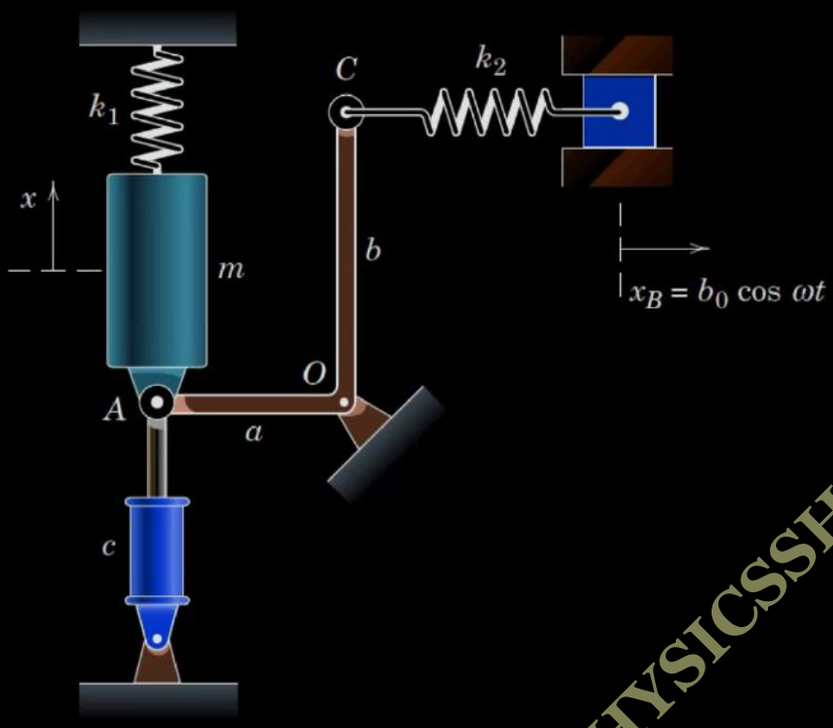
$$\Rightarrow \quad m\ddot{x} + c\dot{x} + kx = kx_B = kb\sin(\omega t)$$

Thus F_0 is replaced by kb. Now rest of the theory that we derived earlier can be applied to solve for $x(t)$.

YOUTUBE.COM/@THEPHYSICSSHELTER

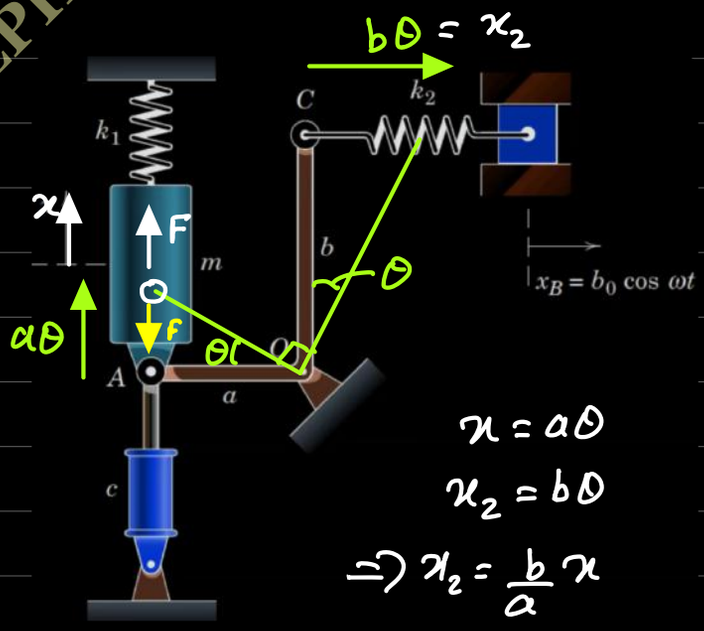
EXAMPLE:

Derive and solve the equation of motion for the mass m in terms of the variable x for the system shown. Neglect the mass of the lever AOC and assume small oscillations.



Solution: Consider the system configuration where pivoted link is tilted through an angle θ, From equilibrium position.

Now net torque on massless link must be zero.



$x = a\theta$

$x_2 = b\theta$

$\Rightarrow x_2 = \frac{b}{a} x$

Let F be the force offered by this link to m. Then by torque balance:

$$Fa = k_2(b_0 \cos \omega t - x_2)b = k_2 b\left(b_0 \cos \omega t - \frac{b}{a} x\right)$$

$$\Rightarrow F = \frac{k_2 b}{a}\left(b_0 \cos \omega t - \frac{b}{a} x\right) = k_2 \mu b_0 \cos \omega t - k_2 \mu^2 x \quad (\mu = b/a)$$

Now Differential equation for vertical motion of m:

$$F - kx - c\dot{x} = m\ddot{x}$$

YOUTUBE.COM/@THEPHYSICSSHELTER

$$\Rightarrow \quad m\ddot{x} + k_1 x + c\dot{x} = k_2 \mu b_0 \cos\omega t - k_2 \mu^2 x$$

$$\Rightarrow \quad m\ddot{x} + c\dot{x} + (k_1 + k_2\mu^2)x = k_2 \mu b_0 \cos\omega t$$

Now this is similar to standard Forced SHMDE :

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t) \qquad \text{Ans.}$$

with $\beta = c/2m$, $\omega_0 = \sqrt{(k_1 + k_2\mu^2)/m}$, $f_0 = k_2 \mu b_0 / m$

YOUTUBE.COM/@THEPHYSICSSHELTER